應用對比學習於水稻坵塊子類別分群與自動標記

Applying Contrastive Learning to the Clustering and Automatic Labeling of Rice Parcel Subclasses

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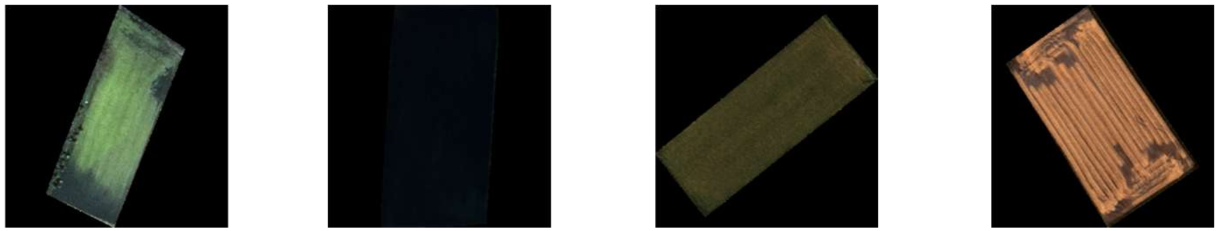
Outline

- Introduction
- Related works
- Research methods
- Experiments
- Conclusions and Future works

Introduction

Background

- Rice is one of the most important staple crops in Taiwan.
- The growth cycle is divided into four main stages: **Transplanting**, **Growing**, **Ripening**, **Harvesting**.
- The Ministry of Agriculture regularly captures **1/5000 aerial images** across Taiwan and has experts label rice subclasses based on image features.
- Each aerial image contains **1,000 to 2,000 parcels**, manual labeling takes around **5 hours per image**.



(a) Transplanting (b) Growing (c) Ripening (d) Harvesting

Fig 1. Rice subclasses

Background

- Supervised learning models have demonstrated high accuracy in crop classification tasks; however, they **still require large amounts of labeled training data**.

- **Contrastive learning** methods (e.g., **SimCLR**, **MoCo**) can adopt a two-stage pipeline to **learn image features** from unlabeled data, followed by **clustering algorithms** (such as K-means) to perform clustering.

- **TURTLE** introduces an **unsupervised transfer learning** approach that leverages multiple pre-trained representation spaces and task encoders for classification tasks.

Motivations & Objectives

- Although the current use of 1/5000 aerial images combined with manual annotation achieves high accuracy, the process is **time-consuming and heavily reliant on expert labor**, making it difficult to scale for large-scale applications.

- Supervised learning models can be trained to achieve high discriminative performance; however, **the cost of annotating such training data** remains a significant bottleneck.

- Recently, **contrastive learning techniques** such as **SimCLR, MoCo**, and the more recent **TURTLE framework** have emerged as viable alternatives. Nevertheless, a potential **misalignment between the feature learning objective and the clustering task** can hinder clustering performance.

- This study aims to apply the SeCu method to the clustering of rice parcel images, with the goal of **improving the accuracy of rice subclasses clustering** and **achieving a low-cost, labor-efficient solution for crop labeling**.

Problem Definition

Given **1/5000 aerial images** without cloud cover and corresponding **shapefiles**.
Develop a more effective automatic clustering system for the subclasses of rice parcels to improve clustering performance with the same number of clusters, ultimately enabling more accurate automatic labeling of rice parcels.
The clustering performance will be evaluated using **ACC, ARI and NMI**, with higher scores indicating better performance.
Input: 1/5000 Aerial image without cloud cover and shapefile of rice parcels.
Output: shapefile with clustering results.

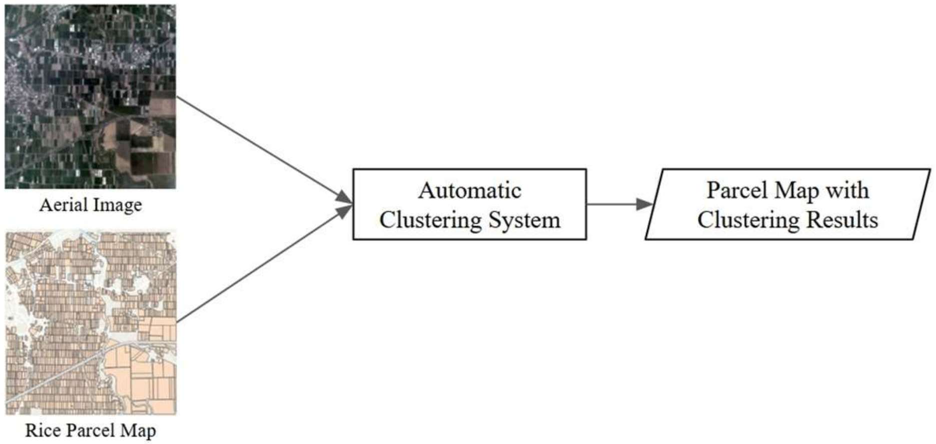


Fig 2. Automatic Clustering System

Contribution

1. Proposed an automatic clustering system combining contrastive learning. Reduced the average annotation time per 1/5000 aerial image **from 5 hours to 10 minutes**. **Achieved 0.8500 clustering accuracy (ACC)** without any labeled data, demonstrating high applicability and potential.

2. Incorporated **Stable Cluster Discrimination (SeCu)** to stabilize unsupervised feature learning. Validated using real Taiwan rice parcel images, showing significant improvements: **ACC: 0.8500, ARI: 0.6927, NMI: 0.7096**, outperforming baseline methods on all three clustering metrics.

3. Investigated the impact of **different cluster numbers ($K \in \{4, 5, 6\}$)**. Found that **$K = 4$ yields the best performance: ARI: 0.6927, NMI: 0.7096**. Highlighted the importance of selecting a suitable number of clusters to ensure clustering accuracy and practicality.

Related works

K-Means

- K-Means aims to **partition data into K clusters** by **minimizing intra-cluster distances** and **maximizing inter-cluster separation**.

- Algorithm Process:
  1. Assigns data points to the nearest cluster centroid.
  2. Updates centroids until convergence.

- Due to its **simplicity and efficiency**, K-Means is **widely used** as a post-processing step **in self-supervised learning pipelines**, to generate initial clustering labels for evaluation or downstream tasks.

SimCLR：Simple Framework for Contrastive Learning of Visual Representations

- SimCLR is a **contrastive learning** framework that learns semantic representations without labels.
- It uses **data augmentation** to create positive pairs (same image, different augmentations) and negative pairs (different images).
- SimCLR uses the Normalized Temperature-scaled Cross Entropy (**NT-Xent**) loss.

$$\mathcal{L}_{i,j} = -\log \frac{\exp\left(\mathrm{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau\right)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp\left(\mathrm{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau\right)}$$

- sim(·): cosine similarity.
- τ: temperature parameter.
- N: batch size.

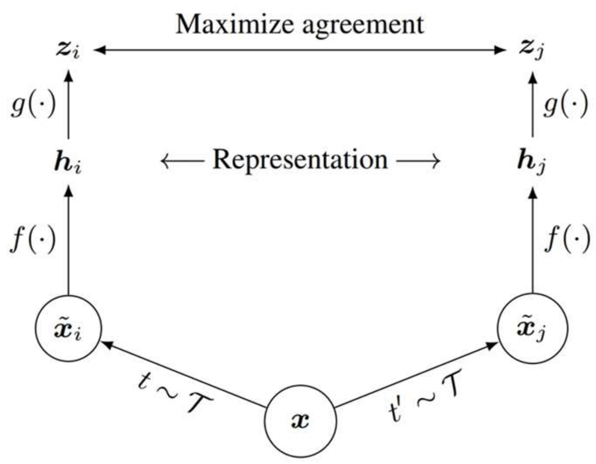


Fig 3. SimCLR

MoCo：Momentum Contrast for Unsupervised Visual Representation Learning

- **Momentum Encoder**: maintains consistency in feature representations.
- **Dynamic Dictionary** (Queue): stores a large number of negative samples, overcoming SimCLR's batch size limitation.
- Loss Function – **InfoNCE**: Similar to SimCLR but operates over a dictionary-based set of negatives.

$$\mathcal{L}q = -\log \frac{\exp(\mathbf{q}\cdot\mathbf{k}^{+}/\tau)}{\sum_{i=0}^{K}\exp(\mathbf{q}\cdot\mathbf{k}_i/\tau)}$$

- τ: temperature parameter.
- k_i: includes k^+ and K negative samples from the queue.
- $q \cdot k$: cosine similarity between normalized vectors.

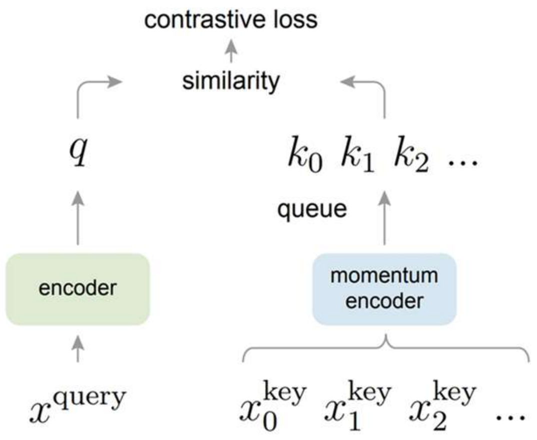


Fig 4. MoCo

Cluster-aware Contrastive Learning for Unsupervised Out-of distribution Detection

- CCL **combines contrastive learning with unsupervised clustering** to enhance both representation quality and clustering stability.

- Incorporates clustering objectives directly into the training process.

- Combines two levels of contrastive signals:
    - **Instance-level contrast**: compares individual augmented samples.
    - **Cluster-level contrast**: encourages representations within the same cluster to be closer.

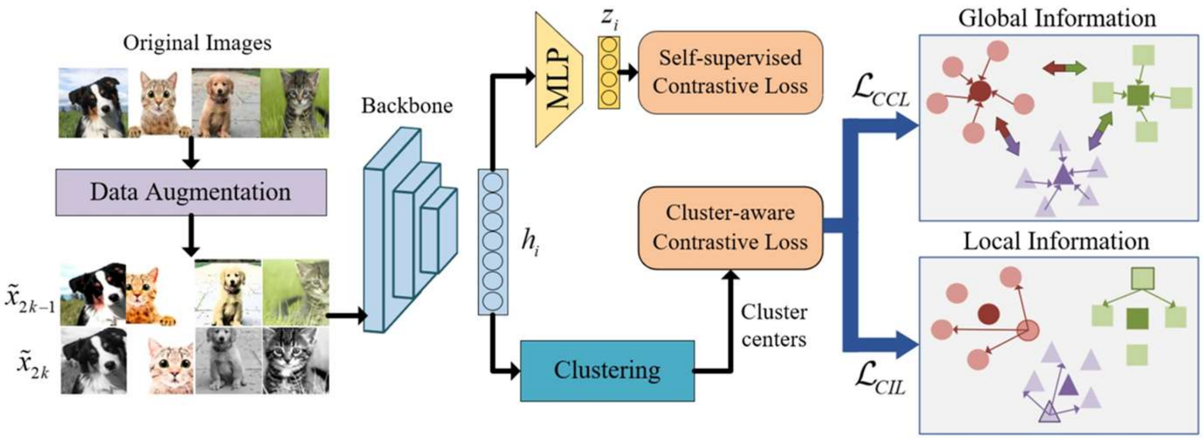


Fig 5. CCL

Stable Cluster Discrimination for Deep Clustering

- Designed for **one-stage deep clustering**, to address instability of standard cross-entropy loss in unsupervised settings.
- **Gradient blocking for negative samples**: Prevents negative samples from influencing cluster center updates, improving stability.
- **Hardness-aware weighting**: Assigns higher weights to hard (ambiguous) samples to improve cluster discriminability.
- **Global entropy constraint**: Encourages balanced cluster assignments to avoid mode collapse.

Let Go of Your Labels with Unsupervised Transfer

- TURTLE introduces an **unsupervised transfer learning** framework that avoids reliance on labeled data.
- **Learns a task encoder** with semantic awareness: Combines **multiple pre-trained feature spaces** to explore and generate semantic classification tasks.

Algorithm 2 TURTLE for Unsupervised Transfer

Require: Dataset D, class count C, total iterations T, representation spaces $\phi_1(\cdot), ..., \phi_K(\cdot)$, task parameters $\theta = \{\theta_1, ..., \theta_K\}$, linear classifiers $w^1, ..., w^K$, learning rate η, optimizer $\Xi(\cdot)$, adaptation steps M, entropy weight γ

Randomly initialize $\theta_1, ..., \theta_K$ and $w_0^1, ..., w_0^K$

for $t = 1$ **to** T **do**

  Sample mini-batch from dataset $X \sim D$

  Generate task from task encoder $\tau_\theta \leftarrow \frac{1}{K} \sum_{k=1}^{K} \sigma(\theta_k \phi_k(X))$

  Update linear classifiers for M steps $\forall k \in [K] : w_M^k \leftarrow \Xi^{(M)}(w_0^k, X)$

  Update task parameters $\forall k \in [K] : \theta_k \leftarrow \theta_k - \eta \frac{\partial}{\partial \theta_k} \left(L_{ce}(w_M^k \phi_k(X), \tau_\theta) + \gamma R(\bar{\tau}_\theta) \right)$

  if warm-start **then**

    update start points $\forall k \in [K], w_0^k \leftarrow w_M^k$

  end if

end for

Output: Task parameters $\theta = \{\theta_1, \ldots, \theta_K\}$

Self-Supervised Backbone Framework for Diverse Agricultural Vision Tasks

- Proposes a **lightweight self-supervised learning framework** tailored for **agricultural image tasks**. Based on **SimCLR**, using **ResNet-50** as the backbone network.

- Explored use cases in various agricultural tasks: Transfer learning, Anomaly detection, Image retrieval, Video preprocessing, Image reconstruction.

- Validates the **scalability and cost-effectiveness** of **self-supervised learning in agriculture**.

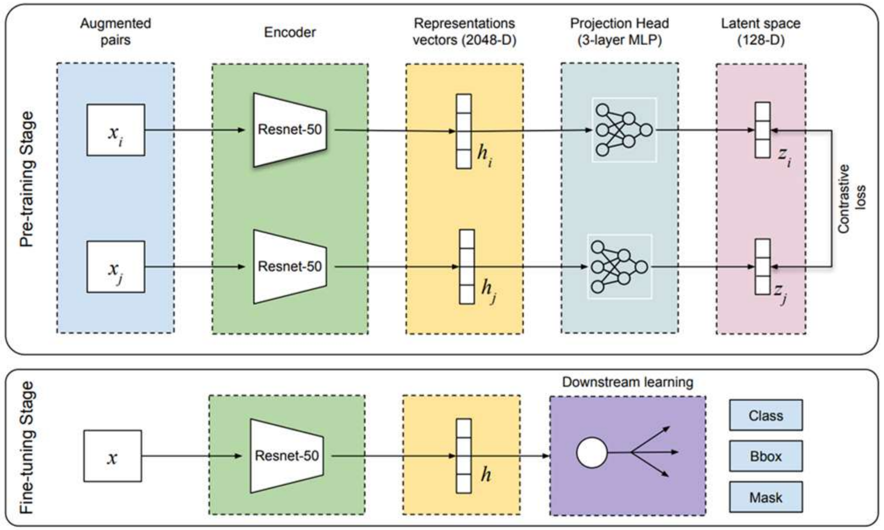


Fig 6. Illustration of the two-stage process

Research Method

Proposed Method

This chapter will be divided into 3 parts:

- Data Preprocessing
- Model Training
- Model Testing

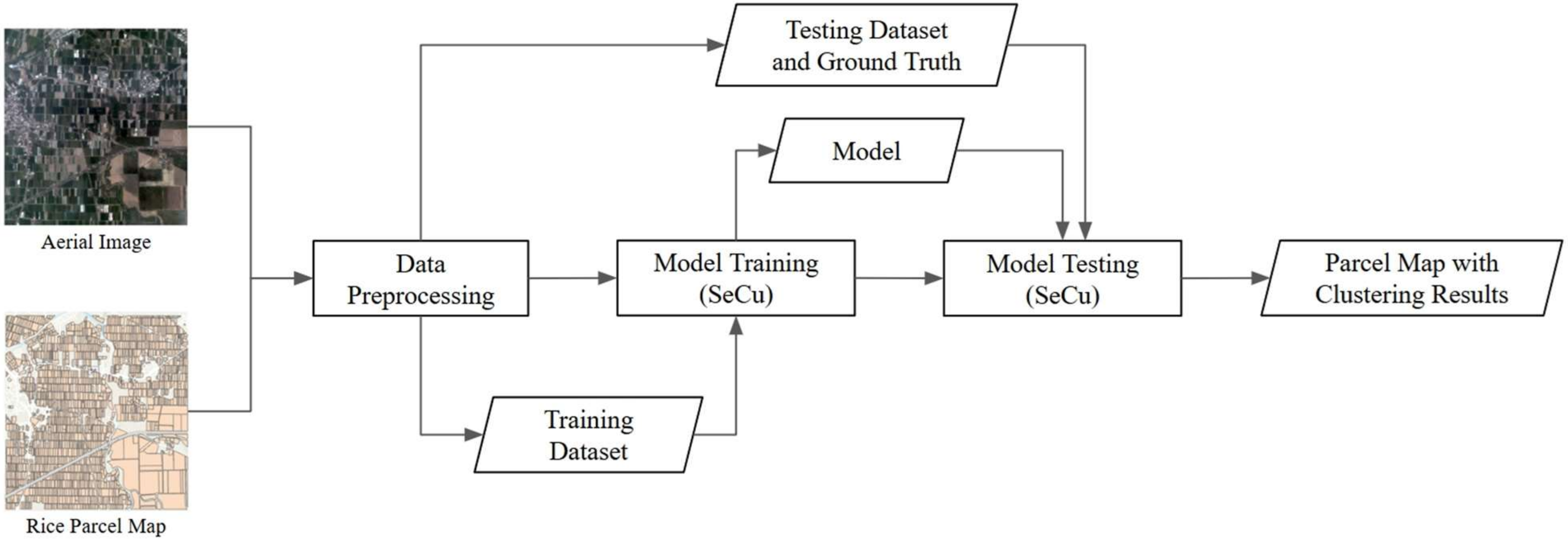


Fig 9. Automatic Clustering System Process

Data preprocessing

Parcel Map	
Content	Polygon shape
Format	Shapefile
Source	Ministry of Agriculture (MOA)

Aerial Image	
Content	Frame id, Shooting date
Format	TIFF
Size	20554 x 22288
Band	R, G, B, NIR
Camera	DMC3
Spatial resolution	1 pixel = 15 x 15 cm

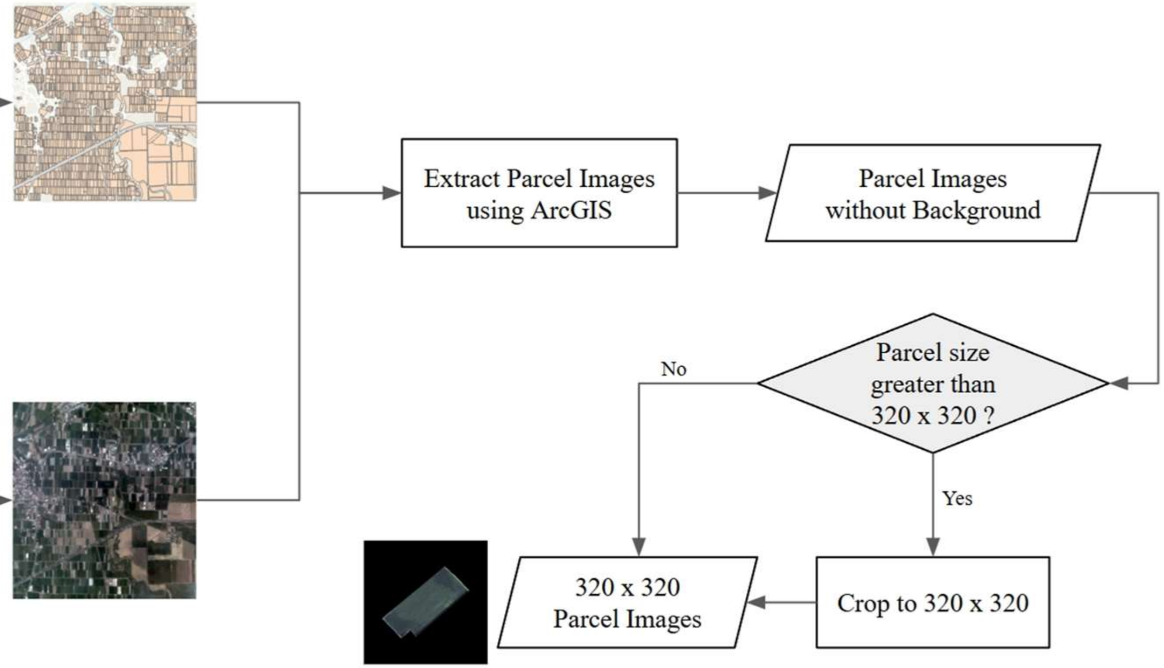


Fig 10. Data Preprocessing

Model Training

We choose to use the **Stable Cluster Discrimination (SeCu)** method to improve the clustering performance:

- Feature Representation Learning
- Cluster Label Update
- Cluster Center Update

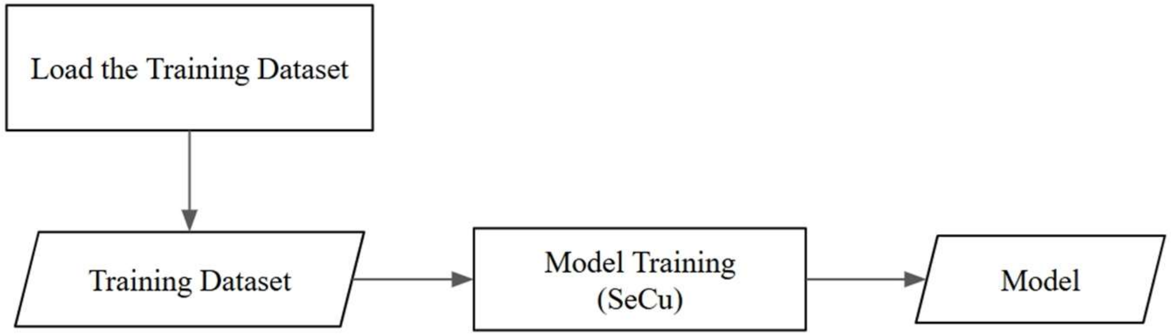


Fig 11. Model Training

Feature Representation Learning

- **Data Augmentation**: For each image, generate two augmented views using: Random resized crop、Color jittering、Random grayscale、Gaussian blur、Horizontal flip
- **Feature Extraction**: Use a shared encoder $f_\theta(\cdot)$ to obtain representations:
  $x_i^1 = f_\theta(\mathrm{aug}_1(i)), \quad x_i^2 = f_\theta(\mathrm{aug}_2(i))$
- **Encoder Architecture**: ResNet-18

$$p_{i,j}^1 = \mathrm{softmax}\left(\frac{(x_i^1)^\top w_j^{t-1}}{\lambda}\right), \quad p_{i,j}^2 = \mathrm{softmax}\left(\frac{(x_i^2)^\top w_j^{t-1}}{\lambda}\right)$$

$$y_i^1 = \tau y_i^{t-1} + (1-\tau)p_i^2, \quad y_i^2 = \tau y_i^{t-1} + (1-\tau)p_i^1$$

$$\ell_{\mathrm{SeCu}}(x_i, y_i) = -\log\left(\frac{\exp(x_i^\top w_{y_i}/\lambda)}{\exp(x_i^\top w_{y_i}/\lambda) + \sum_{k:k\neq y_i}\exp(x_i^\top \tilde{w}_k/\lambda)}\right)$$

$$\mathcal{L}_{\mathrm{rep}} = \sum_{i=1}^{N}\frac{1}{2}\left[\ell_{\mathrm{SeCu}}(x_i^1, y_i^1) + \ell_{\mathrm{SeCu}}(x_i^2, y_i^2)\right]$$

- τ: smoothing coefficient.
- λ: temperature parameter.
- N: total number of training samples in the dataset.
- $\tilde{w}_k$: stop-gradient cluster center for class k.

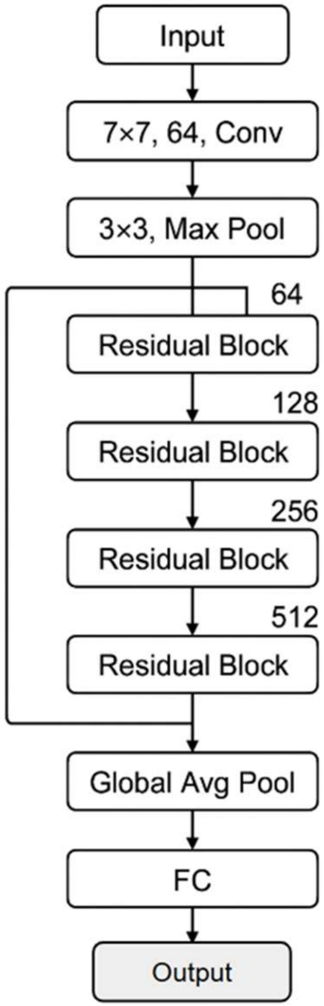


Fig 12. ResNet-18 Architecture

Cluster Label Update

- The latest predicted distribution is calculated using the features extracted by the current encoder and the current cluster center w^t.
- To avoid over-concentration of samples in a single cluster, we use **Size Constraint** to **prevent cluster collapse**.
- The objective is to assign each sample to the cluster with the highest average predicted probability, while ensuring that **each cluster contains no fewer than γN/K samples**.

$$p_{i,j}^1 = \text{softmax}\left(\frac{(x_i^1)^\top \cdot w_j^t}{\lambda}\right), \quad p_{i,j}^2 = \text{softmax}\left(\frac{(x_i^2)^\top \cdot w_j^t}{\lambda}\right)$$

- N: total number of samples.
- K: number of clusters.
- γ: lower bound control parameter.
- ρ_j: dual variable for cluster j.

$$y_{i,j}^t = \begin{cases} 1, & \text{if } j = \arg\min_j \left(-(\log(p_{i,j}^1) + \log(p_{i,j}^2))/2 - \rho_j\right) \\ 0, & \text{otherwise} \end{cases}$$

$$\sum_i y_{i,j} \geq \frac{\gamma N}{K}, \quad \forall j = 1, \ldots, K$$

Cluster Center Update

- **Hardness-aware** Weighting Strategy: Samples with lower prediction confidence (harder samples) receive higher influence on the cluster center.

- Enhances representation of boundary (uncertain) samples, Mitigates overfitting to easy (high-confidence) samples.

$$w_j = \Pi_{\|w_j\|_2=1}\left(\frac{\sum_{i:y_i=j}(1-p_{i,j})x_i}{\sum_{i:y_i=j}(1-p_{i,j})}\right)$$

- $p_{i,j}$: confidence score for assigning x_i to cluster j.
- $\Pi_{\|wj\|2=1}$: projection to the unit sphere to keep consistent length.

Model Testing

Evaluate clustering performance of the trained model on the testing dataset using ground truth labels:

- Model Inference
- Evaluation Metrics Calculation
- Visualization Analysis

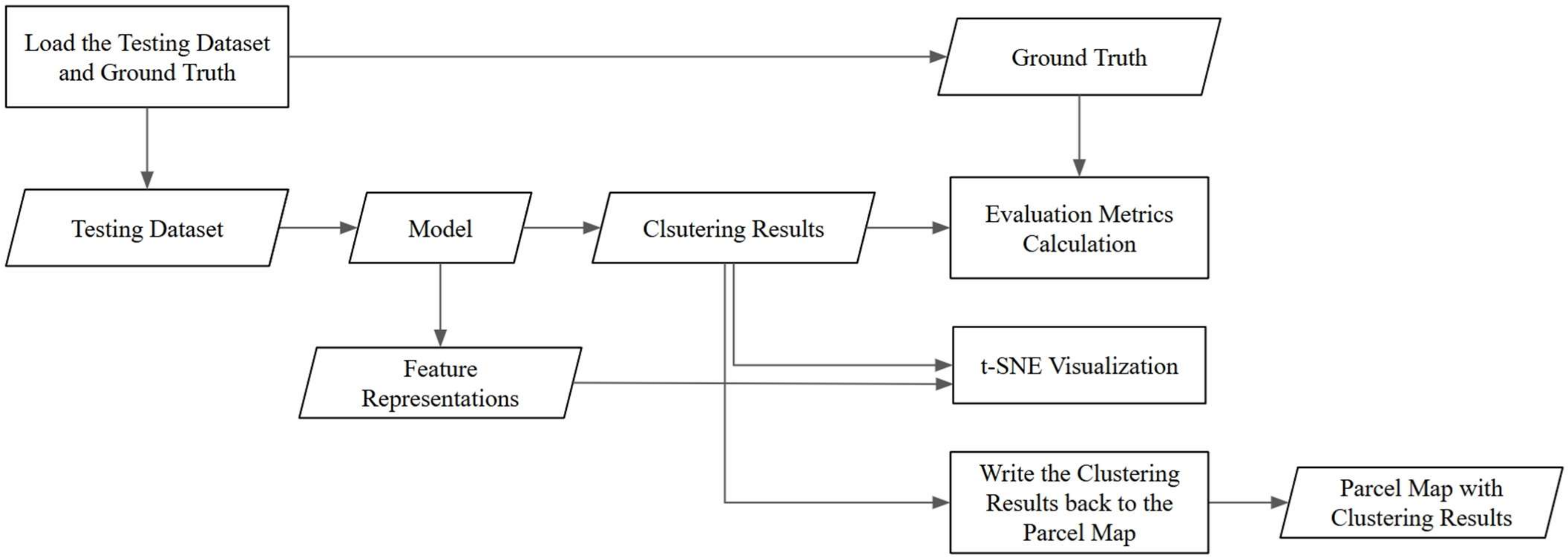


Fig 13. Model Testing

Experiments

Data Introduction

We utilizes a set of 11 aerial images provided by the Ministry of Agriculture. The images were captured in February, April, and June of 2023, covering four counties in Taiwan: Yilan, Taitung, Kaohsiung, and Pingtung.

Table 1. Training Dataset

航照影像編號	水稻坵塊數量	生長階段
水稻_01 插秧期_001_95204077_O230227d	398	插秧期
水稻_02 生長期_002_94171020_O230402b	492	生長期
水稻_02 生長期_003_97223036_O230622c	200	生長期
水稻_02 生長期_004_94182047_O230404a	193	生長期
水稻_03 黃熟期_001_97223028_O230622c	421	黃熟期
水稻_04 收割期_001_96183014_O230623b	319	收割期

Table 2. Testing Dataset

航照影像編號	水稻坵塊數量	生長階段
水稻_01 插秧期_002_95204077_O230227d	256	插秧期
水稻_02 生長期_005_94182047_O230404a	323	生長期
水稻_03 黃熟期_001_97223036_O230622c	166	黃熟期
水稻_03 黃熟期_002_96183014_O230623b	101	黃熟期
水稻_04 收割期_002_96183014_O230623b	174	收割期

Table 3. Aerial image information

Aerial Image	
Content	Frame id, Shooting date
Year	2023
Format	TIF file
Size	20554x22288
Band	R, G, B, NIR
Camera	DMC3
Spatial resolution	1 pixel = 15 x 15 cm
Source	Ministry of Agriculture(MOA)

Evaluation — Clustering Accuracy

- A clustering evaluation metric used to measure the alignment between ground truth labels and clustering results.
- It computes the best label assignment and evaluates the percentage of correctly classified samples.
- Range: ACC values range from [0,1].

Table 4. Consistency of ACC

Value of ACC	Strength of Agreement
0.8 - 1.0	Perfect match
0.6 - 0.8	High similarity
0.4 - 0.6	Moderate Similarity
0.2 - 0.4	Weak Similarity
0 - 0.2	Poor Similarity

$$\mathrm{ACC} = \max_{m} \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}(l_i = m(c_i))$$

- n: Total number of samples.
- c_i: Predicted cluster label for sample i.
- l_i: Ground truth label for sample i.
- $m(\cdot)$: Best mapping function from cluster assignments to ground truth labels, found via the Hungarian algorithm.
- $\mathbb{1}(\cdot)$: Indicator function, which returns 1 if the condition is true, and 0 otherwise.

Evaluation — Adjusted Rand index

- An evaluation metric used to measure the similarity between ground truth labels and clustering results.
- It is an adjusted version of the Rand Index (RI).
- Range: ARI values range from [−1,1].

Table 5. Consistency of ARI

Value of ARI	Strength of Agreement
1	Perfect match
0.5 - < 1	High similarity
0 - < 0.5	Moderate Similarity
0	Random clustering
-1 - < 0	Strong Disagreement

$$\text{ARI} = \frac{\sum_{ij}\binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \Big/ \binom{n}{2}\right]}{\frac{1}{2}\left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}\right] - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \Big/ \binom{n}{2}\right]}$$

- n: Total number of samples.
- n_{ij}: Number of samples that are in predicted cluster i and in ground truth class j.
- $a_i = \sum_j n_{ij}$: Total number of samples in predicted cluster i.
- $b_j = \sum_i n_{ij}$: Total number of samples in ground truth class j.
- $\binom{n}{2}$: Number of possible sample pairs.

Evaluation — Normalized Mutual Information

- An evaluation metric used to measure the similarity between ground truth labels and clustering results.
- It is an extension of Mutual Information (MI).
- Range: NMI values range from [0,1].

Table 6. Consistency of NMI

Value of NMI	Strength of Agreement
0.8 - 1.0	Perfect match
0.6 - 0.8	High similarity
0.4 - 0.6	Moderate Similarity
0.2 - 0.4	Weak Similarity
0 - 0.2	Poor Similarity

$$\mathrm{NMI} = \frac{\sum_{h=1}^{k^{(a)}} \sum_{\ell=1}^{k^{(b)}} n_{h,\ell} \log\left(\frac{n \cdot n_{h,\ell}}{n_h^{(a)} \cdot n_\ell^{(b)}}\right)}{\sqrt{\left(\sum_{h=1}^{k^{(a)}} n_h^{(a)} \log\left(\frac{n_h^{(a)}}{n}\right)\right)\left(\sum_{\ell=1}^{k^{(b)}} n_\ell^{(b)} \log\left(\frac{n_\ell^{(b)}}{n}\right)\right)}}$$

- n: Total number of samples.
- $k^{(a)}$, $k^{(b)}$: Number of clusters in predicted labels and ground truth labels.
- $n_h^{(a)}$: Number of samples in predicted cluster h.
- $n_\ell^{(b)}$: Number of samples in ground truth class ℓ.
- $n_{h,\ell}$: Number of samples that belong to cluster h and class ℓ.

Experiment 1 : Clustering Performance Comparison with Existing Methods

Motivation	This experiment aims to evaluate the performance of the proposed method in the task of rice parcel subclasses clustering and compare it with baseline methods.

- SimCLR + K-Means
- MoCo + K-Means
- TURTLE

Table 7. Training hyperparameter

Hyper-parameter	Configuration
Backbone	ResNet-18
Loss	SeCu
Temperature	0.05
Optimizer	SGD
Learning rate	0.01
Batch Size	64
Epochs	100

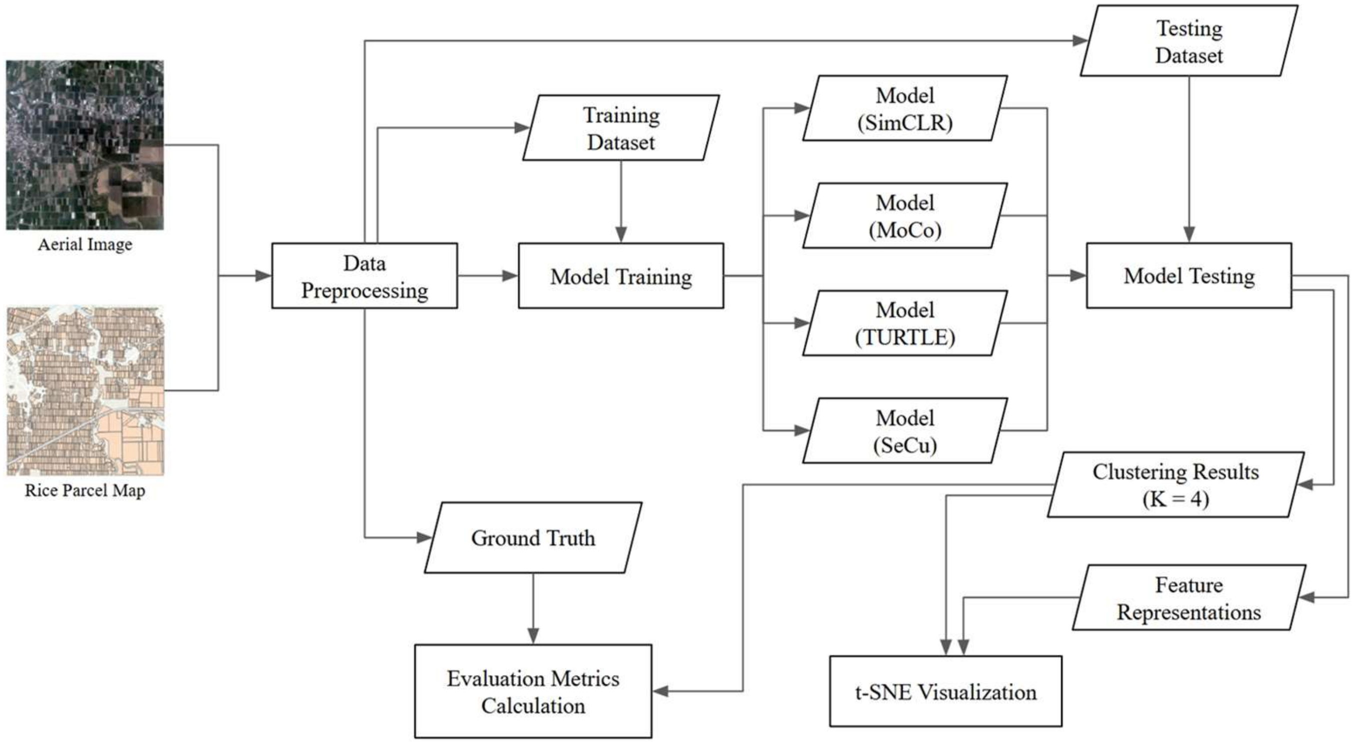


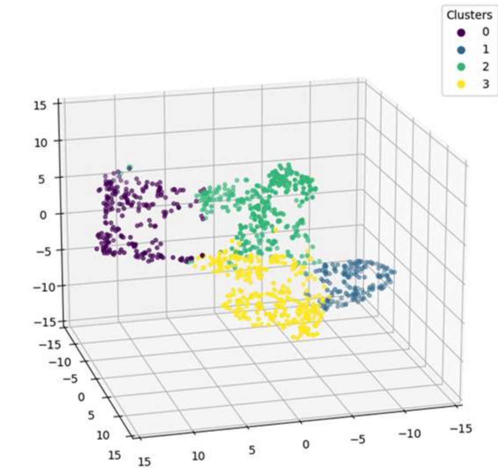
Fig 13. Experiment 1

Experiment 1 Result

Table 8. Clustering performance of each method

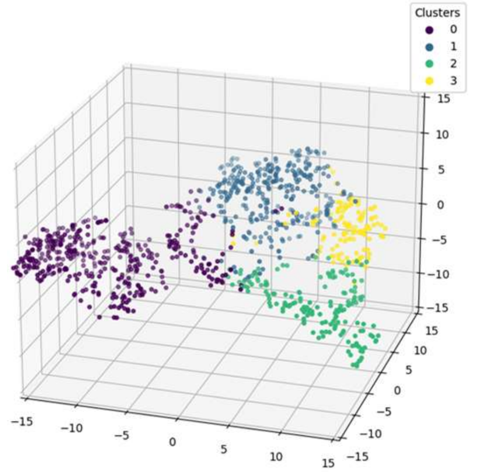
Method	ACC	ARI	NMI
SimCLR + K-Means	0.6951	0.4116	0.5197
MoCo + K-Means	0.6735	0.4272	0.4397
TURTLE	0.7598	0.4921	0.5038
SeCu（Proposed）	**0.8500**	**0.6927**	**0.7096**

SimCLR + K-Means



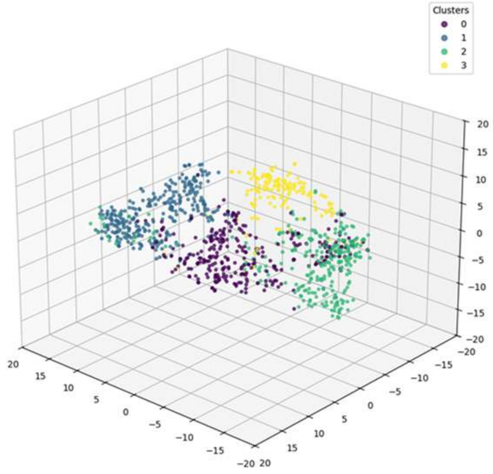
Cluster	坵塊數	插秧期	生長期	黃熟期	收割期
0	221	0%, 0	100%, 221	0%, 0	0%, 0
1	160	2%, 4	0%, 0	1%, 2	96%, 154
2	323	21%, 69	30%, 97	47%, 151	2%, 6
3	316	58%, 183	2%, 5	36%, 114	4%, 14

MoCo + K-Means



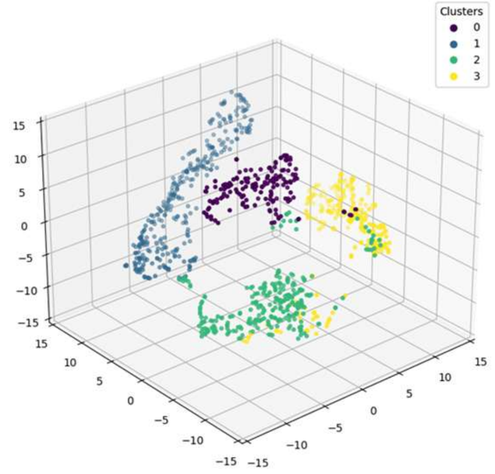
Cluster	坵塊數	插秧期	生長期	黃熟期	收割期
0	409	4%, 17	79%, 323	12%, 50	5%, 19
1	320	22%, 69	0%, 0	33%, 105	46%, 146
2	165	35%, 58	0%, 0	64%, 106	1%, 1
3	126	89%, 112	0%, 0	5%, 6	6%, 8

TURTLE



Cluster	坵塊數	插秧期	生長期	黃熟期	收割期
0	293	69%, 201	10%, 28	19%, 57	2%, 7
1	288	6%, 16	94%, 271	0%, 0	0%, 1
2	291	10%, 30	8%, 22	25%, 73	57%, 166
3	148	6%, 9	1%, 2	93%, 137	0%, 0

SeCu (Proposed)



Cluster	坵塊數	插秧期	生長期	黃熟期	收割期
0	164	0%, 0	0%, 0	**99%, 162**	1%, 2
1	312	0%, 0	**100%, 312**	0%, 0	0%, 0
2	259	**86%, 222**	4%, 11	10%, 25	0%, 1
3	285	12%, 34	0%, 0	28%, 80	60%, 171

Experiment 2 : Analysis of the Impact of Cluster Number on Clustering Performance

Motivation	This experiment aims to investigate how varying the number of clusters affects the performance of the SeCu method, and verify whether setting $K = 4$ best captures the semantic structure of the real-world data.

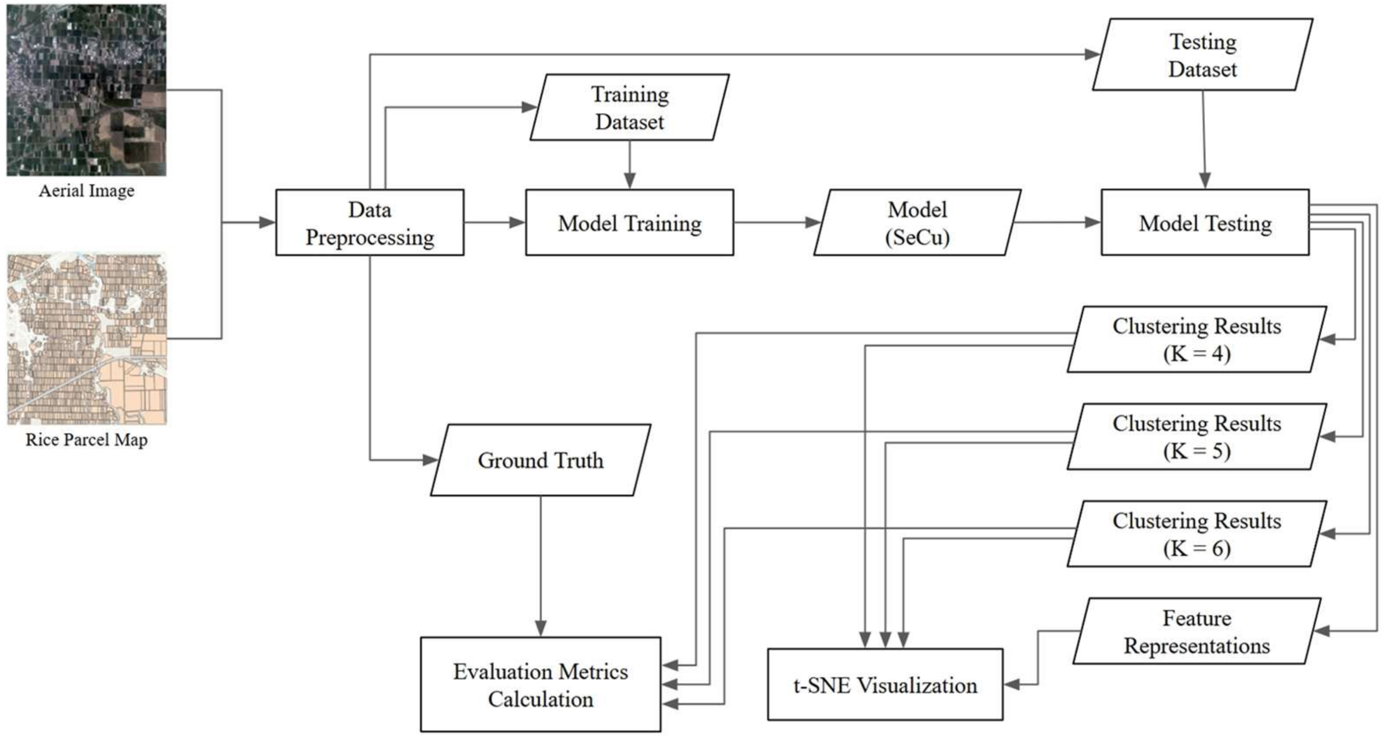


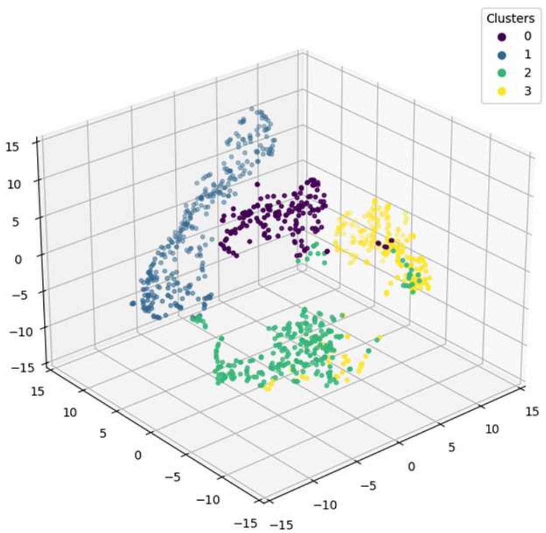
Fig 14. Experiment 2

Experiment 2 Result

Table 9. Clustering performance with different cluster numbers

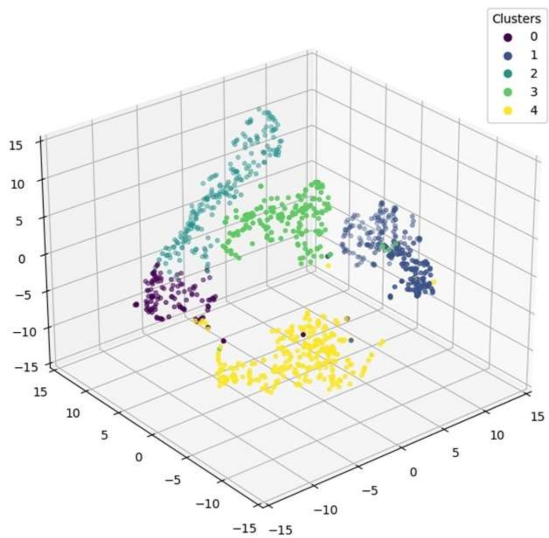
Cluster Number (K)	ARI	NMI
4	**0.6927**	**0.7096**
5	0.6543	0.7430
6	0.6078	0.7045

K = 4



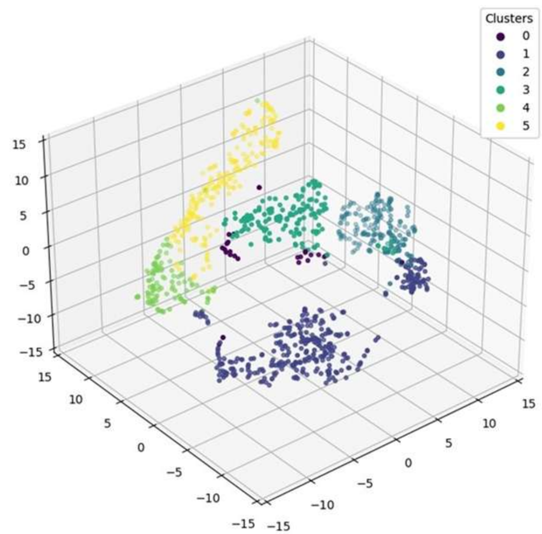
Cluster	坵塊數	插秧期	生長期	黃熟期	收割期
0	164	0%, 0	0%, 0	99%, 162	1%, 2
1	312	0%, 0	100%, 312	0%, 0	0%, 0
2	259	86%, 222	4%, 11	10%, 25	0%, 1
3	285	12%, 34	0%, 0	28%, 80	60%, 171

K = 5



Cluster	坵塊數	插秧期	生長期	黃熟期	收割期
0	113	3%, 3	97%, 110	0%, 0	0%, 0
1	269	1%, 3	0%, 0	35%, 94	64%, 172
2	206	0%, 0	100%, 206	0%, 0	0%, 0
3	170	1%, 1	0%, 0	98%, 163	1%, 6
4	262	95%, 249	3%, 7	2%, 6	0%, 0

K = 6



Cluster	坵塊數	插秧期	生長期	黃熟期	收割期
0	27	11%, 3	0%, 0	81%, 22	7%, 2
1	333	76%, 253	4%, 12	20%, 68	0%, 0
2	198	0%, 0	0%, 0	14%, 27	86%, 171
3	151	0%, 0	0%, 0	99%, 150	1%, 1
4	107	0%, 0	100%, 107	0%, 0	0%, 0
5	204	0%, 0	100%, 204	0%, 0	0%, 0

Conclusions

Conclusions

- Introduced an **automatic clustering method** for **rice parcel images** in Taiwan: Combines **contrastive learning** and **SeCu** to cluster rice parcels into subclasses representing different growth stages, without labeled data.

- Compared with baseline methods, the **proposed method achieved the best results on all three clustering metrics**: ACC、ARI、NMI, t-SNE visualization and category distribution analysis confirmed the semantic separability and intra-cluster consistency of proposed method.

- Reduced the manual annotation time per 1/5000 aerial image **from 5 hours to 10 minutes**, significantly lowering labor cost.

- Experiments showed that **K = 4 yielded the most stable and semantically aligned results**. Over-clustering (e.g., K > 4) led to semantic confusion and class overlap, degrading clustering quality.

Future works

- Extend the proposed method to **other crop types and geographic regions**. Evaluate its generalization ability and stability under varying agricultural environments.

- Some **clusters still show overlap at boundaries**, especially for semantically similar subclasses.

- Future improvements may include:
  1. Enhancing the loss function to increase inter-cluster separation.
  2. Designing contrastive strategies sensitive to borderline samples.

- Cluster ambiguity may stem from **growth-stage transitions** in real crops. Future work can explore **temporal analysis** by integrating multi-timepoint image data to better understand and model these transitions.

Thanks for listening